

Assignment 1 – Entity-Component Framework [Part 1]

Goal

To put in practice object-oriented programming principles and techniques covered in class, including inheritance, use of polymorphism, downcasting and up casting. The goal for this assignment is to Implement an Entity-Component framework for game development using an Entity-Component architecture. This is a primordial building block of modern game architectures, so it is important that all students get some experience with their internals.

It is a common challenge for students to have difficulty expressing design in terms of entities and components. This exercise should provide an opportunity to work on a familiar domain and apply these notions in a guided and controlled environment.

Description

The first part of the assignment focuses on developing a working framework. To do this, students should follow the indications proposed by the test_driver program (test_driver.cpp) and the test suite (test_suite.hh). Upon completion students should end up having implemented a usable framework that can represent the entity-component architecture.

The main goal of this first part is to make all tests compile and pass.

Students must implement the Component class and the Entity class with their respective member attributes and methods, following the next criteria:

- Component class must implement interfaces IComparable, IClonable, IPrintable
- Component class must have an identification attribute (id) of type int. The attribute must be only visible in the context of the class or its children's.
- Entity class can store or attach objects of type Component.
- Component class must have member of type Entity storing its owner. The owner of the Component must be assigned at the moment of attaching it to the Entity.

What to do:

The code explicitly requires the following files, which the student should create and submit:

- component.hh and component.cc
- entity.hh and entity.cc

Submission Guidelines

Submit only DrawLine.cpp with a Readme.txt file as a zip folder on Moodle with the following naming conventions:

cs225sp26_{username}_1.1.zip